

Budget Envelope for the Mechanical Power Module (MPM)

Takeaway: This budget envelope defines the *total financial architecture* required to take the MPM from engineering → pilot → standard → consortium → national deployment. It is structured as a standards-grade budgeting framework, not a line-item spreadsheet, and is designed for grant applications, consortium formation, and institutional planning.

1. Budget Philosophy

The MPM is a universal mechanical-power primitive. Its budget envelope must:

- Fund **open, vendor-neutral standardization**
- Support **multi-vendor manufacturing**
- Enable **pilot deployments** across Colorado and national infrastructure
- Build **swap-station logistics** and transport systems
- Maintain **regulatory alignment** with DOE, FERC, ASME, NIST
- Stand up the **MPM Consortium** as a durable governance institution

This envelope is intentionally modular so it can scale with deployment phases.

2. Budget Envelope Summary

Below is the high-level envelope across the seven operational domains. Each domain begins with a Guided Link so you can expand it.

A. Engineering & Standards

Range: \$2.5M–\$4M Covers geometry definition, interface tolerances, safety envelope, reference implementations, and compliance test suite development.

B. Pilot Fabrication & Host Integration

Range: \$3M–\$6M Funds pilot module production, host-system adapters, instrumentation, and integration labor.

C. Logistics Infrastructure

Range: \$1.5M–\$3M Includes swap stations, transport frames, handling systems, operator training, and site preparation.

D. Pilot Deployment & Telemetry

Range: \$2M–\$4M Supports controlled pilot operations, telemetry infrastructure, data collection, and safety validation.

E. Regulatory Alignment

Range: \$750K–\$1.5M Funds DOE/NREL collaboration, FERC permitting alignment, ASME/NIST harmonization, and compliance documentation.

F. Consortium Formation & Governance

Range: \$1M–\$2M Covers formation of the Standards Board, Compliance Authority, Deployment Council, and competitive-neutrality enforcement.

G. National Scaling & Manufacturing Enablement

Range: \$5M–\$12M Supports multi-vendor manufacturing, national deployment planning, and ISO/IEC harmonization.

3. Total Budget Envelope

Total Estimated Envelope: \$15.75M – \$32.5M This range reflects:

- Multi-vendor manufacturing readiness
- Colorado pilot deployments
- National regulatory alignment
- Consortium formation
- Standards v1.0 publication
- Logistics infrastructure build-out

This is a *deployment-grade* envelope, not a research-only budget.

4. Budget Breakdown by Phase

Phase 1 — Standardization (v0.9 → v1.0)

\$2.5M–\$4M Geometry, interface, safety envelope, reference implementations, compliance suite.

Phase 2 — Pilot Fabrication

\$3M–\$6M Pilot modules, host adapters, instrumentation.

Phase 3 — Logistics Infrastructure

\$1.5M–\$3M Swap stations, transport frames, operator training.

Phase 4 — Pilot Deployment

\$2M–\$4M Duty-cycle testing, telemetry, safety validation.

Phase 5 — Standards Refinement

\$500K–\$1M Data analysis, tolerance refinement, v1.0 publication.

Phase 6 — Consortium Launch

\$1M–\$2M Governance, compliance authority, membership onboarding.

Phase 7 — National Scaling

\$5M–\$12M Manufacturing enablement, regulatory alignment, ISO/IEC harmonization.

5. Budget Envelope by Workstream

Each workstream begins with a Guided Link.

- **Engineering & Standards** — \$2.5M–\$4M
- **Manufacturing & Fabrication** — \$3M–\$6M
- **Logistics & Deployment** — \$1.5M–\$3M
- **Telemetry & Data** — \$1M–\$2M
- **Regulatory & Compliance** — \$750K–\$1.5M
- **Consortium Governance** — \$1M–\$2M
- **Scaling & Manufacturing Enablement** — \$5M–\$12M

6. Budget Envelope Characteristics

The MPM budget envelope is:

- **Modular** — supports phased deployment
- **Vendor-neutral** — avoids proprietary capture
- **Regulator-aligned** — supports DOE, FERC, ASME, NIST
- **Logistics-aware** — funds swap stations and transport frames
- **Standards-driven** — prioritizes geometry, safety, compliance
- **Scalable** — supports national and international adoption

7. Success Criteria

The budget envelope is successful when:

- All pilot modules and host systems are fabricated and deployed
- Swap stations operate with <15-minute module exchange
- Telemetry supports real-time compliance
- MPM Standard v1.0 is published
- Consortium governance is operational
- Multi-vendor manufacturing is enabled
- Regulators recognize the MPM as a non-proprietary standard